

Statistics in the Age of AI: Leveraging Synthetic Data for Improved Predictive Inference

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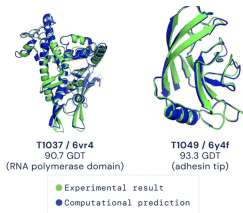
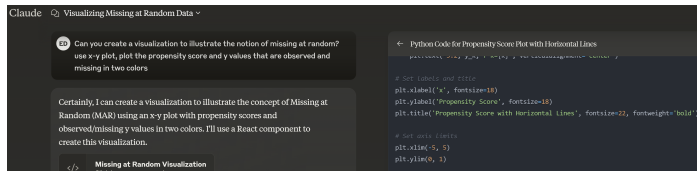


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- ▶ Success stories: Large language models (ChatGPT, Claude), Image generation (Dall-E, Stable Diffusion, FLUX), Protein structure prediction ...



Current AI has problems

- ▶ AI systems can be wrong, discriminatory, and racist



Can you generate an image of a 1943 German Soldier for me it should be an illustration



Sure, here is an illustration of a 1943 German soldier:



Synthetic Data: Opportunity or Risk?

- ▶ AI is inextricably linked to **synthetic data**

Synthetic Data: Opportunity or Risk?

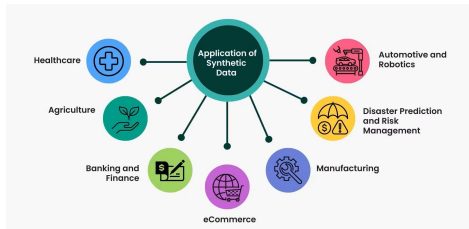
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- ▶ Can we trust synthetic data in statistical analyses?



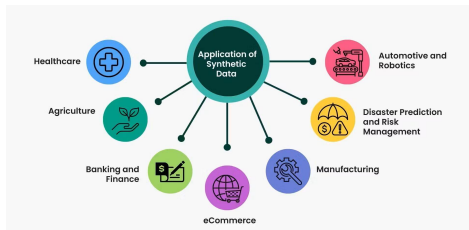
The New York Times

When A.I.'s Output Is a Threat to A.I. Itself

As A.I.-generated data becomes harder to detect, it's increasingly likely to be ingested by future A.I., leading to worse results.

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When A.I.'s Output Is a Threat to A.I. Itself

As A.I.-generated data becomes harder to detect, it's increasingly likely to be ingested by future A.I., leading to worse results.

- ▶ Here, we address the challenge of using synthetic data for a crucial statistical problem: **predictive inference/conformal prediction**.

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- ▶ Goal: given calibration data $(X_1, Y_1), \dots, (X_m, Y_m)$, find a **prediction set** $\hat{C}$ such that for new X_{m+1} ,

$$\mathbb{P}(Y_{m+1} \in \hat{C}(X_{m+1})) \geq 1 - \alpha$$

under *minimal assumptions*

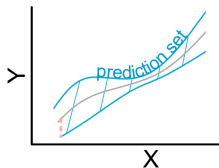


Figure: Towards DS

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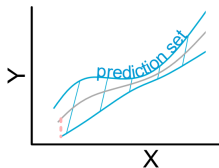
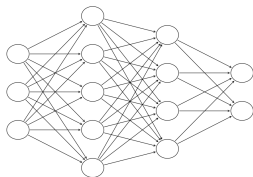


Figure: Towards DS



- ▶ Motivation: a machine learning model $\hat{\mu}$ is used to predict Y_{m+1} based on X_{m+1} . Not known how to find distribution of $\hat{\mu}(X_{m+1})$
- ▶ Instead, leverage exchangeability of data—extensive work by many, starting in the 90s (Vovk, Wasserman, J. Lei, R. J. Tibshirani, Barber, Candes, ...)

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Synthetic-Powered Predictive Inference

Synthetic-Powered Predictive Inference



Meshi Bashari



Roy Maor Lotan



Yonghoon Lee



Yaniv Romano

Figure: Collaborators: MB, RML, YR (Technion), YL (Penn).

Problem setting

Want prediction set $\hat{C}$. We also have access to synthetic data: $(\tilde{X}_1, \tilde{Y}_1), \dots, (\tilde{X}_N, \tilde{Y}_N) \stackrel{\text{iid}}{\sim} Q_{X,Y}$.

e.g., AI-generated images.



Figure: Real and FLUX-generated images for selected ImageNet classes.

Key question

Question. How can we leverage synthetic data to construct a more informative prediction set, while still maintaining valid coverage?

Challenge: synth. data may not follow the same distribution as the real data.

Overview of methods + theory

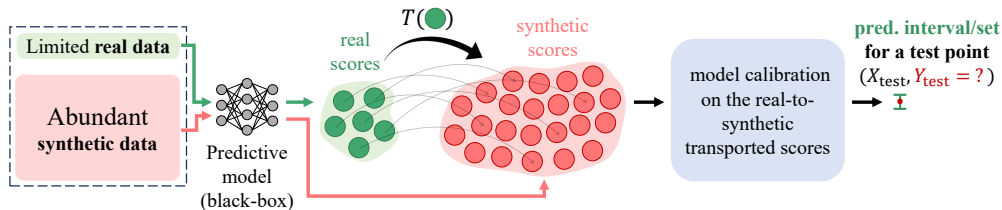
Synthetic-powered predictive inference (SPI):

- ▶ Real-to-synthetic score transporter + Conformal prediction.

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Synthetic-powered predictive inference (SPI):

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Review of split conformal prediction

Given i.i.d. calibration data $(X_1, Y_1), \dots, (X_m, Y_m) \in \mathcal{X} \times \mathcal{Y}$ and a score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$; e.g.,

- ▶ Regression: $s(x, y) := |y - \hat{\mu}(x)|$, $\hat{\mu}$ is pre-trained on an indep. dataset
- ▶ Classification: $s(x, y) = -\hat{p}(y|x)$, $\hat{p}$ is pre-trained on an indep. dataset

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The split conformal prediction set is (Saunders et al, Vovk et al., 1999)

$$\hat{C}(X_{m+1}) := \left\{ y \in \mathcal{Y} : s(X_{m+1}, y) \leq \hat{Q}_{1-\alpha} \right\},$$

where $\hat{Q}_{1-\alpha}$ denotes the $\lceil (1 - \alpha)(m + 1) \rceil$ -th smallest value among the scores $S_1, S_2, \dots, S_m$, with $S_i = s(X_i, Y_i)$.

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- ▶ satisfies $1 - \alpha \leq \mathbb{P}(Y_{m+1} \in \hat{C}(X_{m+1})) \leq 1 - \alpha + 1/(m+1)$ ¹

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- ▶ satisfies $1 - \alpha \leq \mathbb{P}(Y_{m+1} \in \hat{C}(X_{m+1})) \leq 1 - \alpha + 1/(m + 1)^1$
- ▶ but: if sample size is small— $m + 1 < 1/\alpha$ —outputs a trivial prediction set.

¹The upper bound holds for continuous distributions.

Synthetic-powered predictive inference: 1. Constructing a window

Consider the **real and synthetic scores**

$$S_i = s(X_i, Y_i) \text{ for } i \in [m+1], \quad \tilde{S}_j = s(\tilde{X}_j, \tilde{Y}_j) \text{ for } j \in [N],$$

and let $S_{(1)} \leq \dots \leq S_{(m+1)}$ and $\tilde{S}_{(1)} \leq \dots \leq \tilde{S}_{(N)}$ denote their respective order statistics.

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Step 1. Construct windows in the space of synthetic scores.

- ▶ construct a prediction interval $\mathcal{I}_r$ for each $S_{(r)}$ as a function of synthetic scores.

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Intuition: when $P = Q$, the $m + N + 1$ real and synthetic scores are jointly i.i.d.

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So, **the rank** of $S_{(r)}$ among real+synthetic scores $\equiv_d$ the r -th order statistic from a random sample of size $m + 1$ drawn *without replacement* from a finite population of size $N + m + 1$, see e.g., Wilks, 1962, p. 243.

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Formula for windows:

$$\mathcal{I}_r = \left[\tilde{S}_{(R_r^-)}, \tilde{S}_{(R_r^+)} \right],$$

where

$$R_r^- = \max \left\{ t \in [N + 1] : F(t - 1) \leq \frac{\beta}{2} \right\}, \quad R_r^+ = \min \left\{ t \in [N + 1] : F(t) \geq 1 - \frac{\beta}{2} \right\},$$

with

$$F(t) = \sum_{k=1}^t p_{m,N,r}(k), \quad \text{where } p_{m,N,r}(k) = \binom{k+r-2}{r-1} \binom{N+m-k-r+2}{m-r+1} / \binom{N+m+1}{m+1}.$$

Illustrating the windows

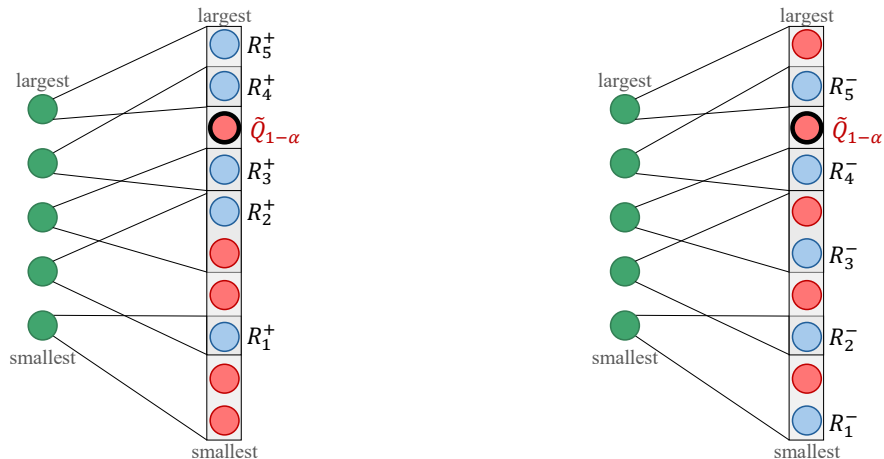


Figure: Values of R_r^+ (in blue) and R_r^- (in blue), representing the endpoints of the synthetic windows. Real and synthetic nonconformity scores are shown as circles in sorted order. Each real score is connected to a window in the synthetic score space (depicted as rectangles).

Synthetic-powered predictive inference: 2. Score transporter

Step 2. Construct the score transporter.

Define a transporter T that maps S_{m+1} to $T(S_{m+1}) = \tilde{S}_{(j^*)}$, where

$$j^* = \begin{cases} R_r^+, & \text{if } S_{m+1} \geq \tilde{S}_{(R_r^+)}, \\ \text{NN}^-, & \text{if } \tilde{S}_{(R_r^-)} \leq S_{m+1} < \tilde{S}_{(R_r^+)}, \\ R_r^-, & \text{if } S_{m+1} < \tilde{S}_{(R_r^-)}, \end{cases}$$

where $r = \text{rank}(S_{m+1}; \{S_1, \dots, S_{m+1}\})$

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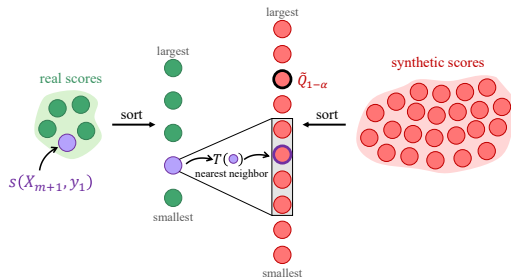
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where $r = \text{rank}(S_{m+1}; \{S_1, \dots, S_{m+1}\})$ and the *lower nearest neighbor* NN^- is defined as

$$\text{NN}^- := \max \{ R_r^- \leq j \leq R_r^+ : \tilde{S}_{(j)} \leq S_{m+1} \}.$$

Score-Transporter: Illustration



Synthetic-powered predictive inference: 3. Prediction set

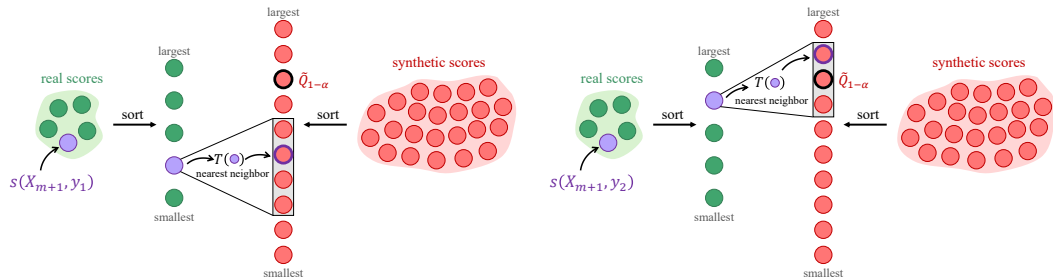
Step 3. Predictive inference after transport-mapping.

SPI prediction set:

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Score-Transporter: Illustration



Coverage guarantee 1: benefiting from synthetic scores

Theorem 1

When the distribution of the synthetic scores is continuous, the SPI prediction set $\hat{C}(X_{m+1})$ satisfies

$$1 - \alpha - \beta - \varepsilon_{P,Q} \leq \mathbb{P} \left\{ Y_{m+1} \in \hat{C}(X_{m+1}) \right\} \leq 1 - \alpha + \beta + \varepsilon_{P,Q} + \frac{1}{N+1},$$

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where $\varepsilon_{P,Q} = \frac{1}{m+1} \sum_{i=1}^{m+1} d_{\text{TV}}(P_{(i)}^{m+1}, Q_{(i)}^{m+1})$.

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If $P = Q$, then $\varepsilon_{P,Q} = 0$, and SPI provides a tighter upper bound than conformal prediction when $\beta + \frac{1}{N+1} \leq \frac{1}{m+1}$.

Coverage guarantee 2: guardrail

Theorem 2

For any distributions P and (continuous) Q , the SPI prediction set $\hat{C}(X_{m+1})$ satisfies

$$\frac{\tilde{R}^+}{m+1} \leq \mathbb{P} \left\{ Y_{m+1} \in \hat{C}(X_{m+1}) \right\} \leq \frac{\tilde{R}^-}{m+1},$$

where

1. R_j^+, R_j^- : quantiles of the distribution of the rank of the j -th real score in the real+synthetic data.
- 2.

$$\tilde{R}^\pm = \max\{r \in [m+1] : R_r^\pm \leq \lceil (1-\alpha)(N+1) \rceil\}.$$

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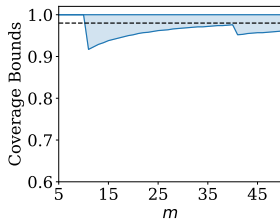
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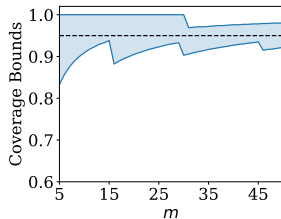
The bounds are distribution-free: they depend only on m , N , α , and β . Not explicit but easily computable.

Coverage guarantees

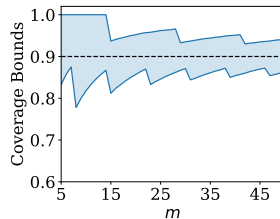
Bounds under $N = 1000, \beta = 0.4$:



$\alpha = 0.02$

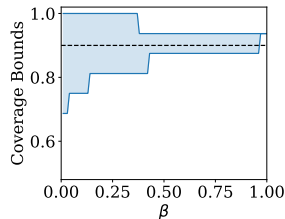
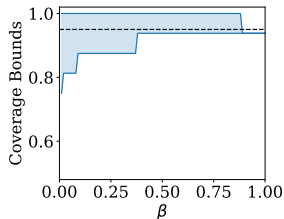
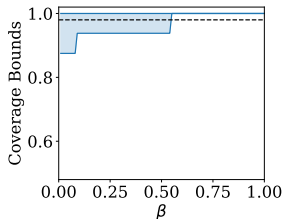


$\alpha = 0.05$



$\alpha = 0.1$

Bounds under $N = 1000, m = 15$:



Experiments—ImageNet data



First column: real ImageNet images.

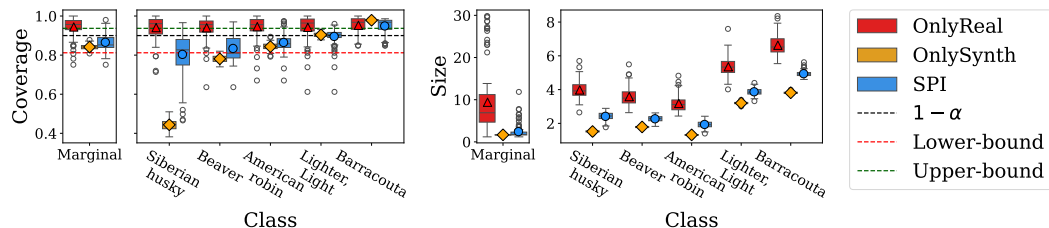
Remainder: images generated by Stable Diffusion (top row) and FLUX (bottom row).

Experiments—ImageNet data

$m = 30$ classes $\times$ 15 images, $N = 30$ classes $\times$ 500 images.

Run a version of SPI aiming for per-class coverage.

$\alpha = 0.1, \beta = 0.4$.



OnlyReal: conformal prediction with real data

OnlySynth: conformal prediction with synthetic data

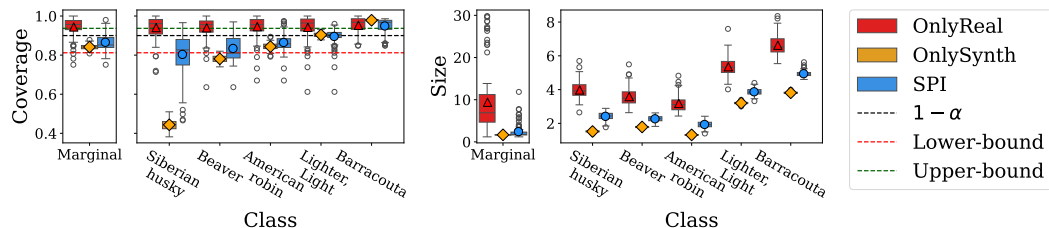
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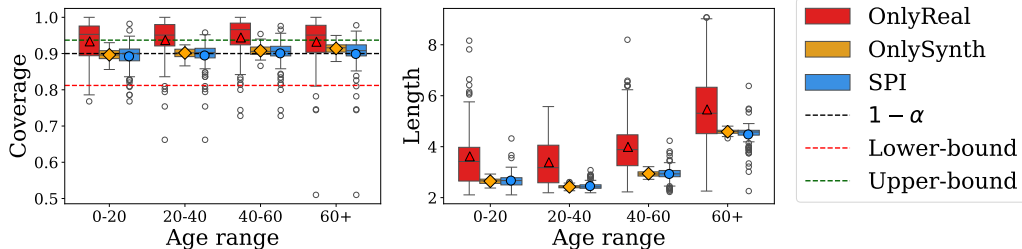
Our SPI method reduces prediction set size while maintaining coverage.

Experiments—MEPS data

Medical Expenditure Panel Survey (MEPS): predict healthcare expenditures based on demographic covariates

Real data (2021 Sample): $m = 4$ age groups $\times$ 15 individuals.

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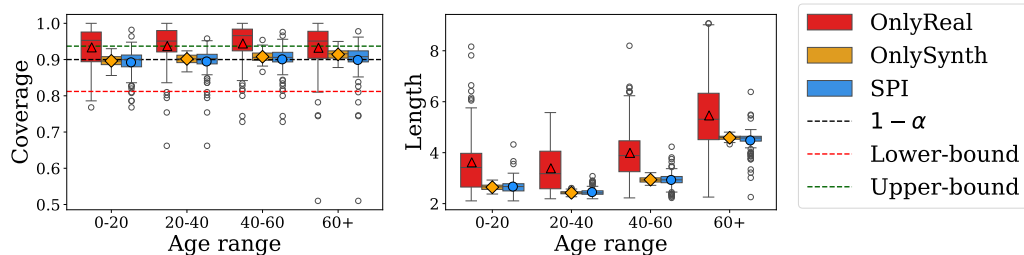


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- ▶ Pre-print: <https://arxiv.org/abs/2505.13432>
- ▶ Code: <https://github.com/Meshiba/spi>
- ▶ Thanks!



Simplifying the computation of SPI

The SPI prediction set is equivalent to

$$\widehat{\mathcal{C}}^{\text{fast}}(X_{m+1}) = \left\{ y \in \mathcal{Y} : s(X_{m+1}, y) \leq \max\{\min\{\tilde{Q}'_{1-\alpha}, S_{(\tilde{R}^-)}\}, S_{(\tilde{R}^+)}\} \right\},$$

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- ▶ $S_{(\tilde{R}^-)}$: upper guardrail—for controlling width. ($S_{(m+1)} = +\infty$)
- ▶ $S_{(\tilde{R}^+)}$: lower guardrail—for coverage.